

- (b) (i) Consider a cylindrical conductor of length l and area of cross-section A . Current I is maintained in the conductor and electrons drift with velocity $\vec{v}_d$ ($|\vec{v}_d| = \frac{e|\vec{E}|}{m} \tau$), (where symbols have their usual meanings). Show that the conductivity σ of the material of the conductor is given by $\sigma = \frac{ne^2}{m} \tau$.
- (ii) The resistance of a metal wire at 20°C is $1.05 \, \Omega$ and at 100°C is $1.38 \, \Omega$. Determine the temperature coefficient of resistivity of this metal.

The amount of charge crossing the area in time Δt is

$$Q = Ne$$

$$Q = (nA l)e$$

$$I = \frac{Q}{\Delta t}$$

$$I = \frac{nA l e}{\Delta t}$$

$$I = nAe |\vec{v}_d| \dots\dots\dots(i)$$



Substituting the value of $|\vec{v}_d|$ in eq.(i)

$$\frac{I}{A} = ne \left(\frac{e |\vec{E}| \tau}{m} \right)$$

$$|\vec{J}| = \frac{ne^2 \tau}{m} |\vec{E}| \dots\dots\dots(ii)$$

where $\vec{J}$ is current density and parallel to $\vec{E}$, and

$$\vec{J} = \sigma \vec{E} \dots\dots\dots(iii)$$

From eq.(ii) and (iii)

$$\sigma = ne^2 \tau / m$$

$$\begin{aligned} (ii) \quad \alpha &= \frac{(R_2 - R_1)}{[R_1 (T_2 - T_1)]} \\ &= \alpha = \frac{(1.38 - 1.05)}{[1.05 \times (100 - 20)]} \\ \alpha &= \frac{0.33}{(1.05 \times 80)} \\ &= 0.0039 \text{ } ^\circ\text{C}^{-1} \end{aligned}$$

1

1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

Alternatively

$$R_1 = R_0 (1 + \alpha \Delta T_1) \dots\dots\dots(i)$$

$$R_2 = R_0 (1 + \alpha \Delta T_2) \dots\dots\dots(ii)$$

$$R_1 / R_2 = \frac{(1 + \alpha \Delta T_1)}{(1 + \alpha \Delta T_2)}$$

$$\frac{1.05}{1.38} = \frac{1 + 20\alpha}{1 + 100\alpha}$$

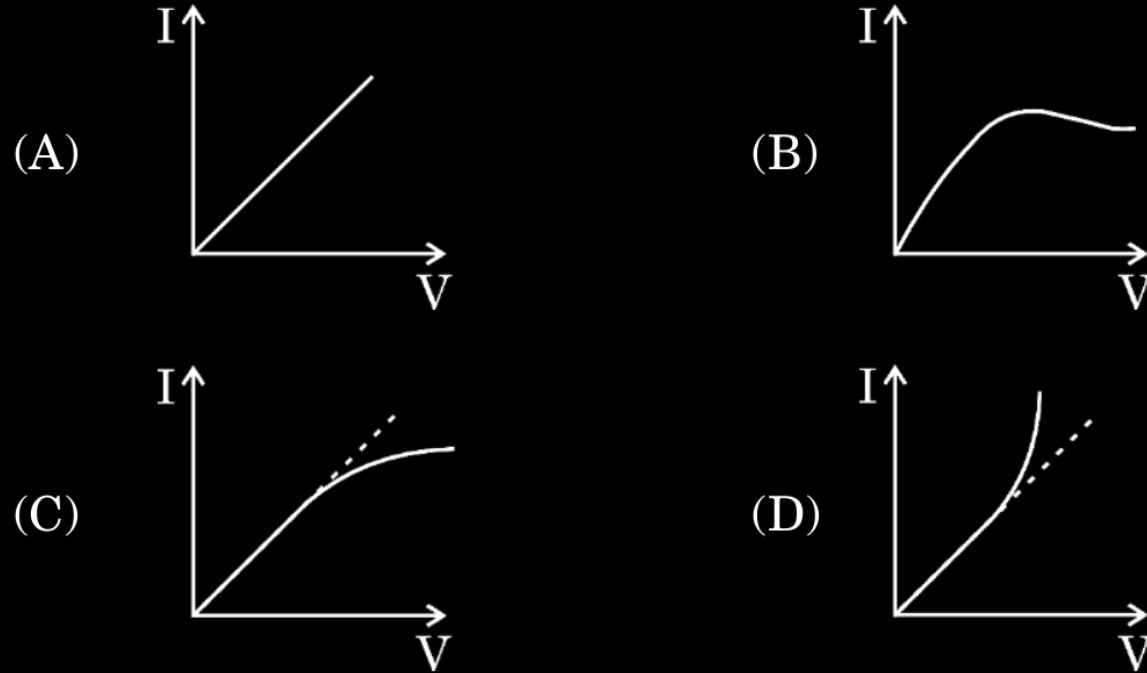
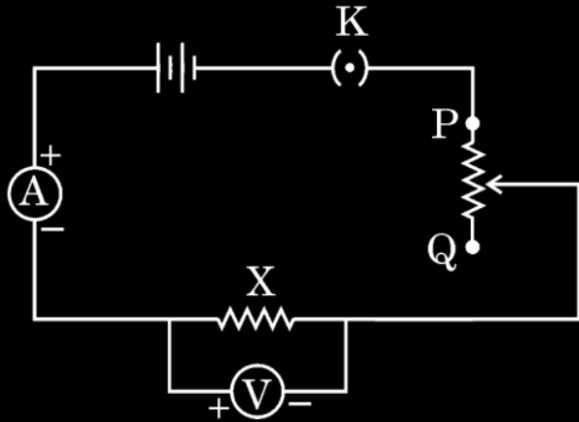
On solving

$$\begin{aligned} \alpha &= \frac{11}{2580} \text{ } ^\circ\text{C}^{-1} \\ &= 0.0042 \text{ } ^\circ\text{C}^{-1} \end{aligned}$$

30. A student sets up the circuit as shown in the figure to find the value of unknown resistance X and records a set of readings of the voltmeter and the ammeter by using the rheostat.

- (i) If resistor X were made of manganin and readings for V and I are taken without switching off the circuit, the graph between V and I will be as :

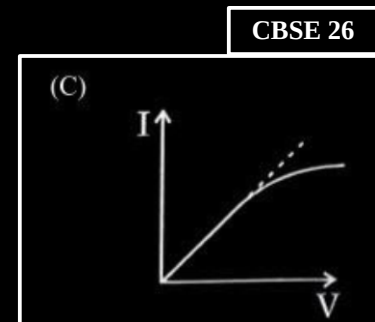
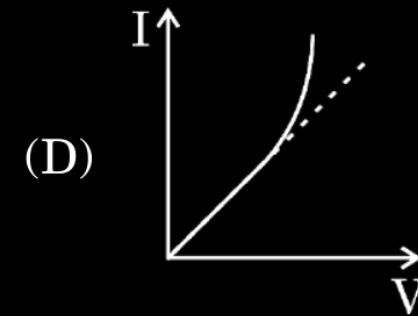
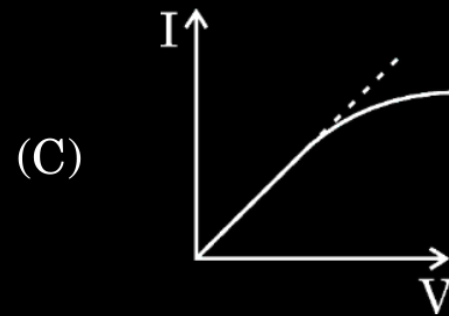
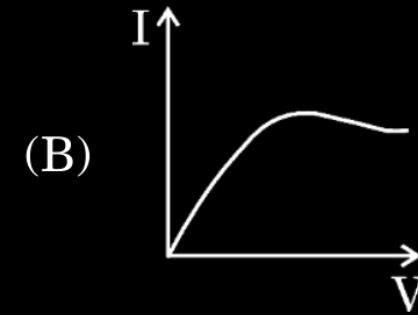
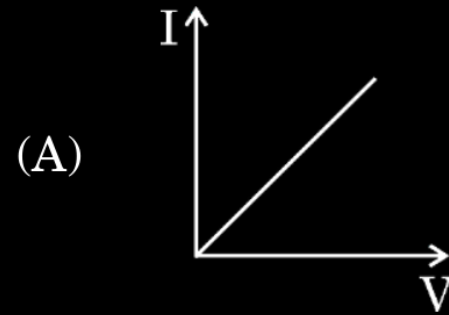
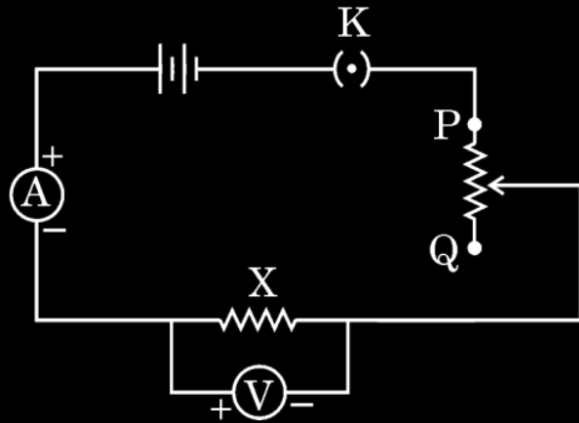
1



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- (i) If resistor X were made of manganin and readings for V and I are taken without switching off the circuit, the graph between V and I will be as :

1



(ii) Error in the value of X obtained from different sets of voltmeter and ammeter readings, is :

1

- (A) due to error in voltmeter reading only.
- (B) due to error in ammeter reading only.
- (C) equal to the sum of error in voltmeter reading and error in ammeter reading.
- (D) equal to error in voltmeter reading divided by the error in ammeter reading.

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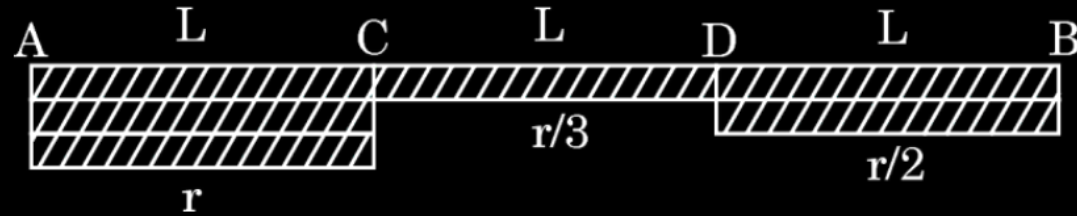
(C) equal to sum of error in voltmeter reading and the error in ammeter reading.
--

- (iii) If the movable end of rheostat is moved towards P, then : 1
- (A) reading in ammeter decreases and reading in voltmeter increases.
 - (B) readings in both voltmeter and ammeter increase.
 - (C) reading in ammeter increases and reading in voltmeter decreases.
 - (D) readings in both voltmeter and ammeter decrease.

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(B) readings in both voltmeter and ammeter increase.

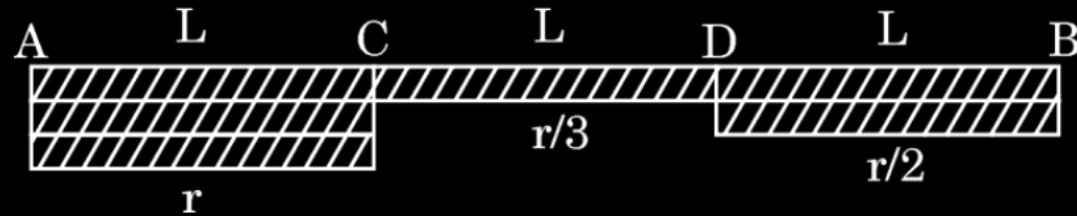
- (iv) (a) Suppose the unknown resistance X is replaced by a wire made of the same metal. This wire consists of three parts, of the same length L but has radii r , $r/3$ and $r/2$ as shown in the figure.



For a particular setting of the rheostat, let v_1 , v_2 and v_3 be the value of drift velocities in parts AC, CD and DB. Then : 1

- (A) $v_1 > v_2 > v_3$
(B) $v_2 > v_3 > v_1$
(C) $v_3 > v_2 > v_1$
(D) $v_1 = v_2 = v_3$

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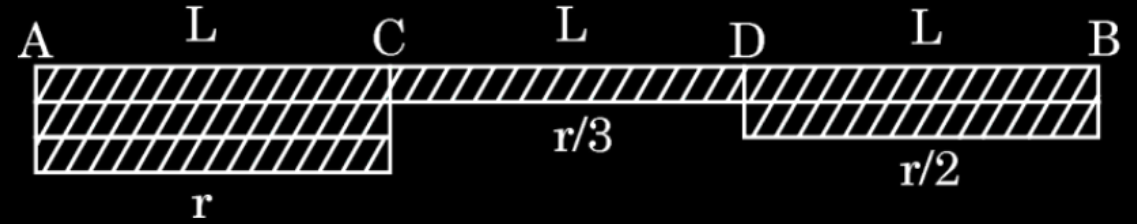
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(B) $v_2 > v_3 > v_1$

(b) Consider the same wire, as shown in figure in question (iv) (a) connected in place of X. For a particular setting of rheostat, let E_1 , E_2 and E_3 be the value of electric fields in part AC, CD and DB. Then :

1

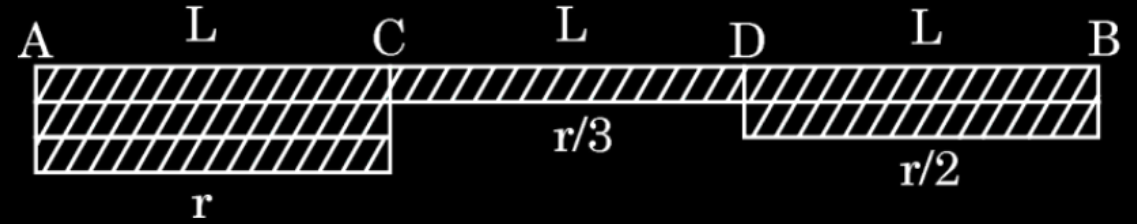
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- (B) $E_3 > E_2 > E_1$
- (C) $E_2 > E_3 > E_1$
- (D) $E_1 > E_2 > E_3$



(C) $E_2 > E_3 > E_1$

(b) (i) Answer the following giving reason :

- (I) The electron drift speed is estimated to be only a few mm/s for currents in the range of a few amperes. How, then, is the current established almost the instant a circuit is closed ?
- (II) A low voltage supply from which one needs high currents must have very low internal resistance. Why ?
- (III) The assertion that $V = IR$ is a statement of Ohm's law is not true. Why ?

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- (II) A low voltage supply from which one needs high currents must have very low internal resistance. Why ?
- (III) The assertion that $V = IR$ is a statement of Ohm's law is not true. Why ?

(i) (I) Electric field is established throughout the circuit, almost instantly (with the speed of light) causing at every point a local electron drift. Establishment of a current does not have to wait for electrons from one end of the conductor travelling to the other end. However, it does take a little while for the current to reach its steady value.	1
(II) $I = \frac{\mathcal{E}}{R + r}$ The maximum current that can be drawn from a cell is for $R=0$ and it is $I_{\max} = \frac{\mathcal{E}}{r}$	1
(III) The Ohm's law asserts that the plot of I vs V is linear i.e. R is independent of V. But this condition is not applicable for diodes.	1

3. Which of the following characteristics of electrons determines the current in the conductor ?

- (A) Drift velocity only
- (B) Thermal velocity only
- (C) Both drift velocity and thermal velocity
- (D) Neither drift nor thermal velocity

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CBSE 26

(A) Drift velocity only

22. (a) (i) An electric field $\vec{E}$ is maintained in a conductor of length L and area of cross-section A . Show that the electrons drift with an average velocity $\vec{v}_d = \frac{-e\vec{E}}{m} \tau$. Here symbols have their usual meanings.
- (ii) Electrons in metals drift due to force experienced by electrons in the electric field inside the conductor. Thus, the motion of electrons should be accelerated. Then why do the electrons acquire a steady average drift speed ?

An electron will suffer collisions with the heavy fixed ions, but after collision it will emerge with the same speed but in random directions.

If we consider all the electrons, their average velocity will be zero since their directions are random. Thus, if there are N electrons and the velocity of the electron at a given time is v_i then:

$$\frac{1}{N} \sum \vec{v}_i = 0$$

$\frac{1}{2}$

When an electric field is present, the electron will be accelerated due to this field by:

$$\vec{a} = \frac{-e\vec{E}}{m}$$

$\frac{1}{2}$

Consider again electron at a given time t. Then its velocity at time t is:

$$\vec{v}_t = \vec{v}_i + \frac{(-e\vec{E})}{m} t_i$$

$\frac{1}{2}$

where $\vec{v}_i$ was the velocity immediately after the last collision and t_i is the time elapsed after the last collision. The average velocity of the electrons at time t is the average of all the v_i .

The collisions of electrons do not occur at regular intervals but at random times. Let so, τ be the electrons average time between successive collisions, known as relaxation time.

Thus, averaging over the N electrons at any given time 't' gives us the average velocity $\vec{v}_d$

$$\vec{v}_d = (\vec{v}_t)_{\text{average}} = (\vec{v}_i)_{\text{av}} - \frac{e\vec{E}}{m} (t_i)_{\text{av}}$$

$$\vec{v}_d = 0 - \frac{e\vec{E}}{m} \tau$$

$\frac{1}{2}$

(ii) Each free electron does accelerate, increasing to its drift speed until it collides with a positive ion of the metal. It loses its drift speed after collision but starts to accelerate and increase its drift speed again only to suffer a collision and so on. On the average, therefore electron acquire only a drift speed.

1

- (a) An electric field $\vec{E}$ is established across the ends of a cylindrical conductor of length L and area of cross-section A . Discuss how electrons attain an average velocity, independent of time. Hence, obtain a relation between current in the conductor and this 'average velocity' of electrons.
- (b) (i) This 'average velocity' is found to be few mm/s for currents in range of a few amperes. How then is current established almost the instant a circuit is closed ?
- (ii) Two copper wires having their radii in the ratio of 3 : 2 are connected in series across a battery. Find the ratio of the drift velocities of the electrons in the wires.

(a) • $\vec{v}_d = \frac{-e\vec{E}\tau}{m}$, where τ is the average time between two successive

collisions, which is constant.

Distance covered by electrons between two successive collisions is almost same. Hence

$$S = \frac{1}{2}at^2 \text{ where } a=\text{constant, } s=\text{almost constant}$$

So t is constant.

Alternatively

At any instant some of the electrons would have spent time more than τ and some less than τ . Thus averaging over all the times we get a constant time. This way we get average velocity independent of time.

- Let n is the number of free electrons per unit volume in the conductor of length L and cross-sectional area A . Total charge crossing area A in time Δt is

$$q = -neA|\vec{v}_d|\Delta t$$

Where negative sign indicates direction of flow of electrons is opposite to direction of electric field.

By definition, electric current is opposite to flow of electrons.

$$\therefore I \Delta t = neA |\vec{v}_d| \Delta t$$

$$I = neA|\vec{v}_d|$$

(b) (i) Electric field is established throughout the circuit, almost instantly causing at every point a local electron drift. As a result the current is established instantaneously.

(ii) $v_d = \frac{I}{ne\pi r^2}$

$$\frac{v_{d1}}{v_{d2}} = \left(\frac{r_2}{r_1}\right)^2$$

$$= \frac{4}{9}$$

20. What is the order of magnitude of drift velocity of electrons in a conductor ?
Deduce the relation between the current flowing through a conductor and drift velocity of electrons in it.

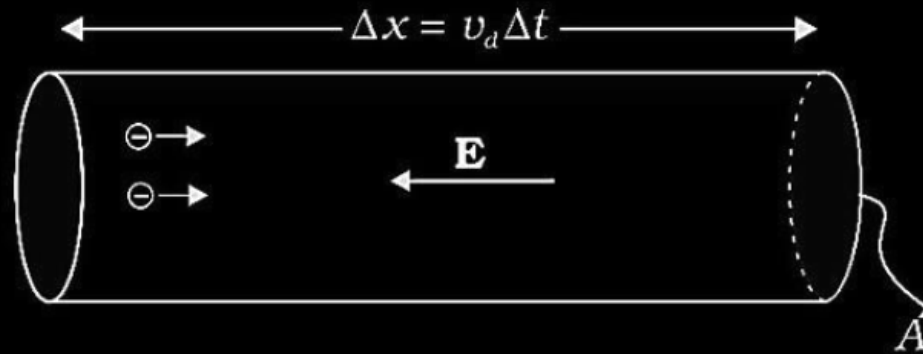
20. What is the order of magnitude of drift velocity of electrons in a conductor ?
Deduce the relation between the current flowing through a conductor and drift velocity of electrons in it.

2

CBSE 26

- Magnitude of drift velocity in conductor is of the order of a few mm/s

$\frac{1}{2}$



- For n number of free electrons per unit volume in a conductor of cross section area A , the total charge transported across the area in time Δt is

$$Q = - ne A v_d \Delta t$$

$\frac{1}{2}$

As electron moves in a direction opposite to that of electric field

$$I \Delta t = ne A v_d \Delta t$$

$\frac{1}{2}$

$$I = ne A v_d$$

$\frac{1}{2}$

- 26.** Name the factors on which electrical conductivity of a material depends. Obtain the relation between current density in a conductor and the conductivity of its material.

3

Factors on which electrical conductivity depends	1½
Obtaining relation between current density and conductivity	1½

Electric conductivity

$$\sigma = \frac{ne^2\tau}{m}$$

Factors on which conductivity depends

(i) concentration of charge carrier

(ii) relaxation time

(iii) mass of charge carrier

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

In a conductor of resistance R , the potential difference V and Current I .

$$V = IR$$

$\frac{1}{2}$

$$El = I\rho \frac{l}{A}$$

$\frac{1}{2}$

$$j = \frac{I}{A} = \frac{E}{\rho}$$

$$j = \sigma E$$

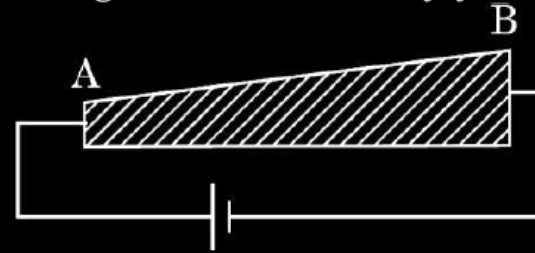
$\frac{1}{2}$

2. A current flows through a series combination of two copper wires of equal length but their radii are in the ratio of 1 : 2. The ratio of drift velocities of free electrons in the wires will be :
- (a) 8 (b) 4 (c) 2 (d) 1

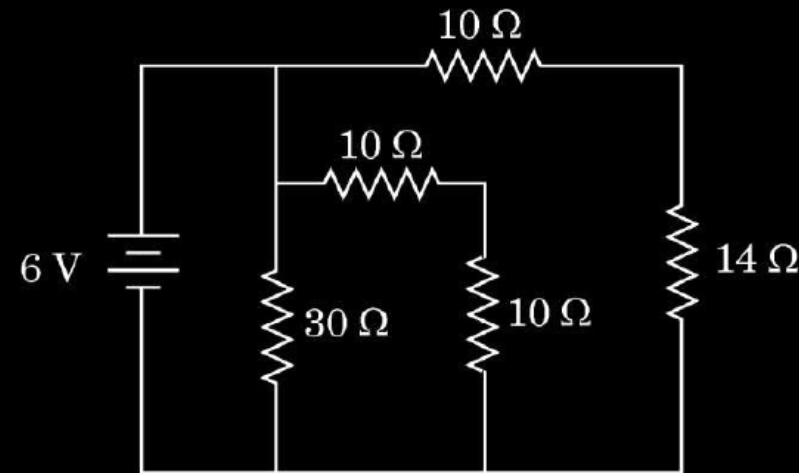
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- (a) 8 (b) 4 (c) 2 (d) 1

(b) 4

32. (a) (i) Define mobility of electrons. Give its SI units.
- (ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire ? Justify your answer.



- (iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.



(i) Definition & S.I. Unit	1+½
(ii) Change in Electric field and drift velocity along the wire	½+½
Justification	½+½
Effective resistance and current	1 ½

(i) Mobility: Mobility is defined as the magnitude of the drift velocity per unit electric field. S.I. Unit: $\frac{m^2}{V.s}$ or $\frac{C.s}{kg}$	1
(ii) Both electric field and the drift velocity decreases. Justification: $v_d = \frac{I}{neA}$ As area increases across the wire, drift velocity decreases. $v_d = \frac{eE}{m} \tau$ As drift velocity decreases, electric field decreases (since e ,m and τ are constant).	$\frac{1}{2} + \frac{1}{2}$
(iii) From the diagram 10Ω and 14 Ω are in series $R_1=10 \Omega + 14 \Omega = 24 \Omega$ 10 Ω and 10 Ω are in series $R_2=10 \Omega+ 10 \Omega = 20 \Omega$ 24 Ω, 20 Ω and 30 Ω are in parallel. $\frac{1}{R} = \frac{1}{24} + \frac{1}{20} + \frac{1}{30} = \frac{5 + 6 + 4}{120} = \frac{15}{120}$ $R= 8 \Omega$ Electric current in the circuit $I = \frac{V}{R} = \frac{6}{8} = \frac{3}{4} A$	$\frac{1}{2}$
Note: Full credit of 1 ½ marks is to be awarded for part (iii) even if a student does not attempt this part.	1

8. The SI unit of mobility of charge carriers is :

(a) $\Omega \text{ s}^{-1}$

(b) $\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$

(c) $\text{m s}^{-1} \text{ V}^{-1}$

(d) $\Omega \text{ m}$

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(a) $\Omega \text{ s}^{-1}$

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(c) $\text{m s}^{-1} \text{V}^{-1}$

(d) $\Omega \text{ m}$

(b) $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$

3. The current density due to drift of electrons in a conductor is given by (symbols have their usual meanings)

(a) $n e A v_d$

(b) $\frac{n A v_d}{e}$

(c) $\frac{n v_d}{e A}$

(d) $n e v_d$

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(d) $n e v_d$

(d) nev_d

26. A potential difference V is applied across a conductor of length l and uniform cross-section area A . How will the (i) electric field E , (ii) drift velocity v_d , and (iii) current density j be affected when (a) V is doubled and (b) l is halved (keeping other factors constant) ?

3

(a) When V is doubled, effect on

(i) Electric field	$\frac{1}{2}$
(ii) Drift velocity	$\frac{1}{2}$
(iii) Current density	$\frac{1}{2}$

(b) When l is halved, effect on

(i) Electric field	$\frac{1}{2}$
(ii) Drift velocity	$\frac{1}{2}$
(iii) Current density	$\frac{1}{2}$

(a) When V is doubled,		
(i) Electric field is doubled.	$\frac{1}{2}$	
(ii) Drift velocity is doubled.	$\frac{1}{2}$	
(iii) Current density is doubled	$\frac{1}{2}$	
(b) When l is halved,		
(i) Electric field is doubled.	$\frac{1}{2}$	
(ii) Drift velocity is doubled.	$\frac{1}{2}$	
(iii) Current density is doubled	$\frac{1}{2}$	

Alternatively: (i) $E = \frac{V}{l}$; $\therefore E' = 2E$ (ii) $v_d = \left(\frac{eE}{m} \right) \tau = \frac{e\tau}{m} \left(\frac{V}{l} \right)$ $v_d \propto V$; $v'_d = 2v_d$ (iii) Current density $J = \frac{I}{A} = \frac{V}{RA}$ $\therefore J$ is also doubled.	When l is halved (i) $E = \frac{V}{l}$; $\therefore E' = 2E$ (ii) $v_d = \left(\frac{eE}{m} \right) \tau = \frac{e}{m} \left(\frac{V}{l} \right) \tau$ $\therefore v'_d = 2v_d$ (iii) $J = \frac{I}{A} = \frac{V}{RA} = \frac{V}{A} \left(\frac{A}{\rho l} \right) = \frac{V}{\rho l}$ $\therefore J$ is also doubled.
---	--

27. Define current density and relaxation time. Derive an expression for resistivity of a conductor in terms of number density of charge carriers in the conductor and relaxation time.

3

Definition of

Current density - $\frac{1}{2}$

Relaxation time - $\frac{1}{2}$

Derivation for resistivity of a conductor - 2

Current density is defined as the current flowing per unit area of cross section of a conductor.

Alternatively

Give full credit if a candidate writes $j=I/A$ in place of definition

Relaxation time is the average time interval between two successive collisions for drifting electrons in a conductor.

From $I = nAev_d$

$$\text{but } v_d = \frac{eE}{m}\tau$$

$$\therefore I = nAe \cdot \frac{eE}{m}\tau$$

$$j = \frac{I}{A} = \frac{ne^2E\tau}{m}$$

$$\text{But } j = \frac{E}{\rho}$$

$$\therefore \rho = \frac{m}{ne^2\tau}$$

1/2

1/2

1/2

1/2

1/2

24. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal.

2

Showing the equal drift velocity in the two conductors (2)

24. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal.

2

$$I = neAVd$$

$\frac{1}{2}$

$$v_d = \frac{I}{neA} = \frac{VA}{neA\rho l}$$

$\frac{1}{2}$

$$v_d = \frac{V}{\rho l ne}$$

$\frac{1}{2}$

In parallel drift velocity is independent of area cross section

So drift velocities are same in both the conductors

$\frac{1}{2}$

Showing the equal drift velocity in the two conductors (2)

33. (a) (i) Explain how free electrons in a metal at constant temperature attain an average velocity under the action of an electric field. Hence obtain an expression for it. 3
- (ii) Consider two conducting wires A and B of the same diameter but made of different materials joined in series across a battery. The number density of electrons in A is 1.5 times that in B. Find the ratio of drift velocity of electrons in wire A to that in wire B. 2

- | | |
|---|-----|
| (i) Explanation | (1) |
| Obtaining expression for average velocity | (2) |
| (ii) Finding ratio of drift velocities | (2) |

(a) (i) Under the effect of external field, an electron experiences a force $\vec{F} = -e\vec{E}$ between collisions.

Due to this force the electron is accelerated and attains a velocity. This velocity is different for different electrons, which averaged over all electrons gives average drift velocity. This drift velocity is constant for a given temperature.

Expression of average velocity:

Under the action of an electric field electrons get accelerated with

$$a = -\frac{eE}{m}$$

Velocity of an electron at any instant of time is

$$\vec{V}_i = \vec{v}_i - \frac{e\vec{E}}{m}t_i$$

Average velocity of the electrons at time ‘t’ is the drift velocity

$$\vec{v}_d = (\vec{V}_i)_{average}$$

$$(\vec{V}_i)_{average} = (\vec{v}_i)_{average} - \frac{e\vec{E}}{m}(t_i)_{average}$$

But $(\vec{v}_i)_{average} = 0$ due to randomness

$$\vec{v}_d = 0 - \frac{e\vec{E}}{m}\tau$$

$$\vec{v}_d = -\frac{e\vec{E}}{m}\tau$$

1

½

½

½

½

(ii) $v_d = \frac{I}{enA} = \left(\frac{4I}{e\pi D^2}\right) \times \frac{1}{n}$	½
$v_d \propto \frac{1}{n}$ for same diameter and current	½
$n_A = 1.5n_B$ (Given)	
$\frac{v_{dA}}{v_{dB}} = \frac{n_B}{n_A} = \frac{2}{3}$	1

24. A potential difference (V) is applied across a conductor of length 'L' and cross-sectional area 'A'.

How will the drift velocity of electrons and the current density be affected if another identical conductor of the same material were connected in series with the first conductor ? Justify your answers.

2

Effect on drift velocity and current density	$(\frac{1}{2} + \frac{1}{2})$
Justification	$(\frac{1}{2} + \frac{1}{2})$

Drift velocity of electrons is halved

$$v_d = \frac{V}{neAR}$$

When another identical conductor of same material were connected in series with the given conductor then total resistance becomes double. Therefore

$$v_d' = \frac{v_d}{2}$$

Current density is halved

$$J = \frac{I}{A} = \frac{V}{RA}$$

$$J' = \frac{J}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

24. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor ? Justify your answers.

2

- | | | |
|-----|--|-----------------|
| (i) | Effect of potential difference on mobility of electron | $(\frac{1}{2})$ |
| | Justification | $(\frac{1}{2})$ |
| (i) | Effect of potential difference on current density | $(\frac{1}{2})$ |
| | Justification | $(\frac{1}{2})$ |

24. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor ? Justify your answers.

2

(i) No effect	$\frac{1}{2}$
$\mu = \frac{v_d l}{V} = \frac{l}{enAR} = \text{Constant}$	$\frac{1}{2}$
Alternatively: $\mu = \frac{v_d}{E} = \frac{e\tau}{m} = \text{Constant}$	
(ii) As $J \propto V$, current density gets doubled	$\frac{1}{2}$
$J = \frac{I}{A} = \frac{V}{RA}$	$\frac{1}{2}$

- | | | |
|-----|--|-----------------|
| (i) | Effect of potential difference on mobility of electron | $(\frac{1}{2})$ |
| | Justification | $(\frac{1}{2})$ |
| (i) | Effect of potential difference on current density | $(\frac{1}{2})$ |
| | Justification | $(\frac{1}{2})$ |

- 26.** A cylindrical conductor of resistance R is connected to a battery. The drift velocity of electrons in this conductor is v_d . The conductor is disconnected from the battery and gradually stretched so that its length increases by 25%. It is again connected to the same battery. Find the new value of : 3
- (a) the drift speed of electrons, and
 - (b) the resistance of the conductor.

(a) Calculation of new value of drift speed	$1\frac{1}{2}$
(b) Calculate of new value of resistance of conductor	$1\frac{1}{2}$

(a)

$$V_d = \left(\frac{eV}{ml}\right)\tau$$

$\frac{1}{2}$

$$V'_d = \left(\frac{eV}{ml'}\right)\tau$$

$\frac{1}{2}$

$$l' = 1.25l$$

$$V'_d = \left(\frac{V_d}{1.25}\right) = \frac{4}{5}V_d$$

$\frac{1}{2}$

(b)

$$A \times l = A' \times l'$$

$\frac{1}{2}$

$$l' = 1.25l$$

$$A' = \frac{A}{1.25}$$

$$R' = \rho \frac{l'}{A'} = \rho \frac{1.25 \times 1.25l}{A}$$

$\frac{1}{2}$

$$R' = \frac{25}{16}R$$

$\frac{1}{2}$

- 32.** (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.
- (ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1.5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be $\left(\frac{1}{2}\right)$ A and $\left(\frac{1}{3}\right)$ A respectively, calculate the internal resistance of each battery.

5

OR

(a) (i) Derivation of relation between I and v_d	2
Explanation	1
(ii) Calculating internal resistance of each battery	2

Total charge transported across the area A in time Δt is

$$\Delta Q = -neAV_d \Delta t \quad \text{-----(1)}$$

Also the amount of charge crossing area 'A' in time Δt is

$$\Delta Q = I \Delta t \quad \text{-----(2)}$$

Comparing equation (1) and (2)

$$I = neAV_d$$

With increase in temperature, average speed of electrons increases resulting in more frequent collisions

Hence relaxation time τ decreases

$$\text{As } R = \frac{ml}{ne^2 \tau A}$$

Resistance increases.

$$\text{(ii) For series } I = \frac{E}{R + r}$$

$$\frac{1}{2} = \frac{3}{R + 2r}$$

$$R + 2r = 6 \quad \text{-----(1)}$$

$$\text{For parallel } \frac{1}{3} = \frac{1.5}{R + \frac{r}{2}}$$

$$2R + r = 9 \quad \text{-----(2)}$$

After solving $r = 1 \Omega$

$\frac{1}{2}$

$\frac{1}{2}$

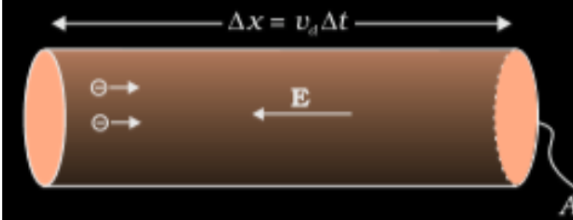
$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

1



28. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2 . When current of 4.25 A is passed through the conductor, calculate
- (i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28} \text{ m}^{-3}$).

(a) Calculating the drift speed

$1\frac{1}{2}$

(b) Calculation of Relaxation time

$1\frac{1}{2}$

28. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2 . When current of 4.25 A is passed through the conductor, calculate
 (i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28} \text{ m}^{-3}$).

(i) $v_d = \frac{I}{enA}$	$\frac{1}{2}$
$= \frac{4.25}{1.6 \times 10^{-19} \times 8.5 \times 10^{28} \times 10^{-6}} \text{ m/s}$	$\frac{1}{2}$
$= 3.125 \times 10^{-4} \text{ m/s}$	$\frac{1}{2}$
(ii) $\tau = \frac{v_d m l}{e V}$	$\frac{1}{2}$
$= \frac{3.12 \times 10^{-4} \times 9.1 \times 10^{-31} \times 5}{1.6 \times 10^{-19} \times 1} \text{ m/s}$	$\frac{1}{2}$
$= 88.72 \times 10^{-16} \text{ s}$	
$= 8.872 \times 10^{-15} \text{ s}$	$\frac{1}{2}$

(a) Calculating the drift speed

$1\frac{1}{2}$

(b) Calculation of Relaxation time

$1\frac{1}{2}$

14. **Assertion (A)** : When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.
Reason (R) : The drift velocity is superposed over large random velocities of electrons.

14. **Assertion (A)** : When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.
Reason (R) : The drift velocity is superposed over large random velocities of electrons.

1

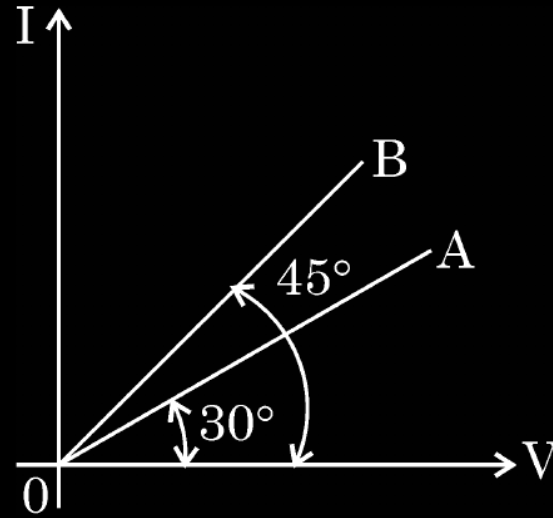
(A) Both assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

20. Two wires A and B of different metals have their lengths in ratio 1 : 2 and their radii in ratio 2 : 1 respectively. I-V graphs for them is shown in the figure. Find the ratio of their

2

(i) Resistances (R_A/R_B) and

(ii) Resistivities (σ_A/σ_B)



(i) Finding resistance $\left(\frac{R_A}{R_B} \right)$ 1

(ii) Finding resistivity $\left(\frac{\sigma_A}{\sigma_B} \right)$ 1

$$(i) \text{ Slope of } I\text{-}V \text{ graph} = \left(\frac{\Delta I}{\Delta V} \right) = \frac{I}{R}$$

$$\frac{R_A}{R_B} = \frac{\text{Slope of B}}{\text{Slope of A}}$$

$$= \frac{\tan 45^\circ}{\tan 30^\circ}$$

$$\frac{R_A}{R_B} = \sqrt{3}$$

(ii)

$$\frac{\sigma_A}{\sigma_B} = \frac{R_A \frac{A_A}{l_A}}{R_B \frac{A_B}{l_B}}$$

$$= \frac{R_A}{R_B} \cdot \frac{A_A}{A_B} \cdot \frac{l_B}{l_A}$$

$$= \sqrt{3} \times \frac{4}{1} \times \frac{2}{1}$$

$$= 8\sqrt{3}$$

1/2

1/2

1/2

1/2

17. (a) What is meant by 'relaxation time' of free electrons in a conductor ?

Show that the resistance of a conductor can be expressed by

$$R = \frac{m l}{n e^2 \tau A}, \text{ where symbols have their usual meanings.} \quad 2$$

OR

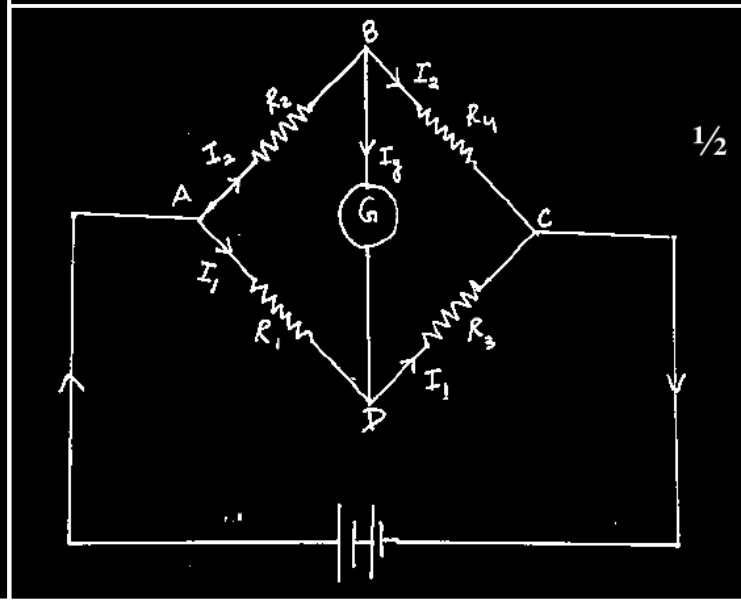
(b) Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it. 2

Meaning of relaxation time	1/2
Derivation of R	1 1/2

Circuit diagram of Wheatstone bridge	1/2
Obtaining the condition when current flows through galvanometer	1 1/2

Average time between two successive collisions of electron in presence of electric field.	1/2
Drift velocity of an electron	
$v_d = \frac{eE}{m} \tau \quad \text{---- (i)}$	1/2
Current flowing through a conductor of length l and area of cross section A	
$I = neAv_d \quad \text{---- (ii)}$	
$I = \frac{ne^2 AE \tau}{m} = \frac{ne^2 A \tau V}{ml}$	1/2
$R = \frac{V}{I} = \frac{ml}{ne^2 \tau A}$	1/2

By applying Kirchoff's loop rule to closed loops ADBA and CBDC	
$-I_1 R_1 + 0 + I_2 R_2 = 0 \quad \text{----- (i) } [I_g = 0]$	1/2
$I_2 R_4 + 0 - I_1 R_3 = 0 \quad \text{----- (ii)}$	
From eq (i) -	
$\frac{I_1}{I_2} = \frac{R_2}{R_1}$	1/2
From eq (ii) -	
$\frac{I_1}{I_2} = \frac{R_4}{R_3}$	
Hence,	
$\frac{R_2}{R_1} = \frac{R_4}{R_3}$	1/2



22. A current of 1.6 A flows through a wire when a potential difference of 1.0 V is applied across it. The length and cross-sectional area of the wire are 1.0 m and $1.0 \times 10^{-7} \text{ m}^2$ respectively. Calculate :

3

(a) Electric field across the wire

(b) Current density

(c) Average relaxation time (τ)

(Number density of free electrons in the wire is $9.0 \times 10^{28} \text{ m}^{-3}$)

Calculation of

(a) Electric field across the wire

1

(b) Current density

1

(c) Average relaxation time (τ)

1

(a) $E = \frac{V}{l}$	$\frac{1}{2}$
$= \frac{1.0V}{1.0m} = 1.0 \text{ V/m}$	$\frac{1}{2}$
(b) $J = \frac{I}{A}$	$\frac{1}{2}$
$J = \frac{1.6A}{1.0 \times 10^{-7} m^2} = 1.6 \times 10^7 \text{ A/m}^2$	$\frac{1}{2}$
(c) $\tau = \frac{m}{ne^2} \frac{J}{E}$	$\frac{1}{2}$
$= \frac{9.1 \times 10^{-31} \times 1 \times 1.6}{9 \times 10^{28} \times (1.6 \times 10^{-19})^2}$	
$= 6.31 \times 10^{-14} \text{ s}$	$\frac{1}{2}$

2. Electrons drift with speed v_d in a conductor with potential difference V across its ends. If V is reduced to $\left(\frac{V}{2}\right)$, their drift speed will become :

(A) $\frac{v_d}{2}$

(B) v_d

(C) $2 v_d$

(D) $4 v_d$

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(A) $\frac{v_d}{2}$

(B) v_d

(C) $2 v_d$

(D) $4 v_d$

(A) $\frac{v_d}{2}$

17. (a) “The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor.” How then is current established almost the instant a circuit is closed ? Explain.
- (b) ‘ $V = IR$ is a statement of Ohm’s Law’ is not true. Explain.

17. (a) “The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor.” How then is current established almost the instant a circuit is closed ? Explain.

(b) ‘ $V = IR$ is a statement of Ohm’s Law’ is not true. Explain.

2

(a) Electric field is established throughout the circuit, almost instantly. It causes a local electron drift at every point, thus establishment of current does not have to wait for electrons from one end of the conductor to travel to other end.	1
(b) Ohm’s law asserts that the plot of I versus V is linear i.e. R is independent of V , while equation $V=IR$ defines resistance and it may be applied to all conducting devices whether they obey Ohm’s law or not.	1

17. Define resistivity of a conductor. How does the resistivity of a conductor depend upon the following :

2

- (a) Number density of free electrons in the conductor (n)
- (b) Their relaxation time (τ)

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2

- (a) Number density of free electrons in the conductor (n)
- (b) Their relaxation time (τ)

Defining resistivity	1	
Dependence of resistivity on		
(a) Number density of free electron	$\frac{1}{2}$	
(b) Relaxation time	$\frac{1}{2}$	
Resistance offered by a material of unit length and having unit cross-sectional area is called resistivity.		1
$\rho = \frac{m}{ne^2\tau}$		
(a) $\rho \propto \frac{1}{n}$		$\frac{1}{2}$
(b) $\rho \propto \frac{1}{\tau}$		$\frac{1}{2}$

- 23.** (a) Define current density. Is it a scalar or a vector ? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m , charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\frac{ne^2}{m} \right) \tau$.

3

OR

- (b) What is a Wheatstone bridge ? Obtain the necessary conditions under which the Wheatstone bridge is balanced.

3

- | | |
|--------------------------------------|---------------|
| • Defining current density | $\frac{1}{2}$ |
| • Whether scalar or vector | $\frac{1}{2}$ |
| • Showing $\vec{j} = \alpha \vec{E}$ | 2 |

- | | |
|--------------------------------|---|
| Defining Wheatstone bridge | 1 |
| Obtaining balancing conditions | 2 |

Current density is the amount of charge flowing per second per unit area normal to the flow.

Alternatively:

$$\vec{j} = \frac{I}{A}$$

It is a vector quantity.

The amount of charge crossing the area A in time Δt is $I \Delta t$, where I is the magnitude of the current. Hence,

$$I \Delta t = ne A |v_d| \Delta t$$

$$I \Delta t = \frac{e^2 A}{m} \tau n \Delta t |E|$$

$$I = |j|A$$

$$|j| = \frac{ne^2}{m} \tau |E|$$

$$\vec{j} = \alpha \vec{E}$$

$\frac{1}{2}$

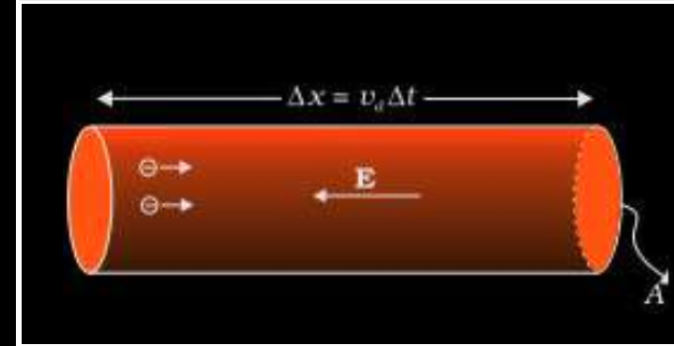
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



22. (a) Define the terms – ‘temperature coefficient of resistance’ and ‘electrical conductivity’ of a conductor. Why are constantan and manganin used for making standard resistances ? Explain. 3

OR

- (b) Define ‘drift velocity’. Show that resistivity of the material of a conductor is inversely proportional to the relaxation time for the free electrons in the conductor. 3

Definition of (i) Temperature coefficient of resistance	1
(ii) Electrical Conductivity	1
Explanation of choosing constantan and manganin for making standard resistances.	1

Definition of Drift velocity	1
Showing resistivity is inversely proportional to relaxation time	2

Drift velocity is the average velocity acquired by electrons under the application of external electric field.

Current flowing through a conductor,

$$I = neAv_d \dots \dots \dots (1)$$

$$v_d = \frac{eE}{m} \tau \dots \dots \dots (2)$$

Substituting (2) in (1)

$$I = \frac{ne^2EA}{m} \tau$$

$$\frac{I}{A} = J = \frac{ne^2E}{m} \tau$$

$$\therefore \sigma = \frac{ne^2}{m} \tau$$

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2 \tau}$$

1

1/2

1/2

1/2

1/2

28. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case.

28. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case.

3

(i) Variation of electric field and justification	$\frac{1}{2} + \frac{1}{2}$
(ii) Variation of current density and justification	$\frac{1}{2} + \frac{1}{2}$
(iii) Variation of mobility of electrons and justification	$\frac{1}{2} + \frac{1}{2}$

With the decrease in area of cross-section.

(i) $E = \frac{I}{A} \rho$	, electric field increases	$\frac{1}{2} + \frac{1}{2}$
(ii) $j = \frac{I}{A}$	, current density increases	$\frac{1}{2} + \frac{1}{2}$
(iii) $\mu_e = \frac{e\tau}{m}$	, mobility remains same	$\frac{1}{2} + \frac{1}{2}$

22. (a) Define mobility of free electrons in a conductor. How does the mobility of electrons in a metal change when :

(i) temperature of the conductor is decreased at constant potential difference ?

(ii) applied potential difference is doubled at a constant temperature ?

3

OR

(b) Obtain an expression for emf and internal resistance of a cell which is equivalent to parallel combination of two cells of emf and internal resistance (E_1, r_1) and (E_2, r_2), respectively.

3

• Definition of mobility	1
• Change in mobility when	
(i) Temperature of conductor is decreased	1
(ii) Applied voltage is doubled	1

Obtaining expression for equivalent emf	2
Equivalent resistance	1

Mobility of free electrons in a conductor is defined as the magnitude of the drift velocity per unit electric field.

1

Alternatively:-

$$\mu = \frac{|\vec{v}_d|}{E}$$

(i) When the temperature of the conductor is decreased at constant potential difference the mobility will increase.

1

(ii) When the applied potential difference is doubled at a constant temperature the mobility will be unchanged.

1

Alternatively:-

$$(i) \quad \mu = \frac{e\tau}{m}$$

As temperature decreases τ will increase and μ will also increase.

$$(ii) \quad \mu = \frac{e\tau}{m}$$

As μ is independent of applied voltage hence μ remains unchanged.

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and $2I$, respectively. Find the ratio of drift velocity of free electrons in the wires.

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and $2I$, respectively. Find the ratio of drift velocity of free electrons in the wires.

2

For the first wire

$$v_d = \frac{I}{enA}$$

$$v_d = \frac{I}{en\pi r^2} \text{-----(1)}$$

For the second wire

$$v'_d = \frac{2I}{\pi en \left(\frac{r^2}{4} \right)} = \frac{8I}{\pi en r^2} \text{-----(2)}$$

From eq (1) and (2)

$$\frac{v'_d}{v_d} = \frac{8}{1}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

29. In a metallic conductor, an electron, moving due to thermal motion, suffers collisions with the heavy fixed ions but after collision, it will emerge out with the same speed but in random directions. If we consider all the electrons, their average velocity will be zero. When an electric field is applied, electrons move with an average velocity, known as drift velocity (v_d). The average time between successive collisions is known as relaxation time (τ). The magnitude of drift velocity per unit electric field is called mobility (μ).

An expression for current through the conductor can be obtained in terms of drift velocity, number of electrons per unit volume (n), electronic charge ($-e$), and the cross-sectional area (A) of the conductor. This expression leads to an expression between current density ($\hat{j}$) and the electric field ($\vec{E}$). Hence, an expression for resistivity (ρ) of a metal is obtained. This expression helps us to understand increase in resistivity of a metal with increase in its temperature, in terms of change in the relaxation time (τ) and change in the number density of electrons (n).

- (i) (a) Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

- (A) I decreases and II increases.
(B) I increases and II is almost constant.
(C) Both I and II increase.
(D) I decreases and II is almost constant.

OR

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- (D) I decreases and II is almost constant.

OR

(b) A steady current flows in a copper wire of non-uniform cross-section. Consider the following three physical quantities :

I. Electric field

II. Current density

III. Drift speed

Then at the different points along the wire :

1

(A) II and III change, but I is constant.

(B) I and II change, but III is constant.

(C) I and III change, but II is constant.

(D) All I, II and III change.

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1

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(B) I and II change, but III is constant.

(C) I and III change, but II is constant.

(D) All I, II and III change.

(iv) Consider two cylindrical conductors A and B, made of the same metal connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is :

1

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

(iv) Consider two cylindrical conductors A and B, made of the same metal connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is :

1

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

17. Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2 . They are connected together with a cell of voltage V . Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel.

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2

(i) Series	$\frac{1}{2}$
$I = neAv_d$	$\frac{1}{2}$
$\frac{v_{d1}}{v_{d2}} = \frac{A_2}{A_1}$	$\frac{1}{2}$
(ii) Parallel	$\frac{1}{2}$
$v_d = \frac{V}{RneA} = \frac{V}{\rho lne}$	$\frac{1}{2}$
$\frac{v_{d1}}{v_{d2}} = \frac{1}{1}$	$\frac{1}{2}$

18. (a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.
- (b) A copper wire of uniform cross-sectional area carries a current of 3.4 A. The drift velocity of conduction electrons is 0.2 mm/s. If the number density of electrons in copper is $8.5 \times 10^{28} \text{ m}^{-3}$, find the area of cross-section of the wire.

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2

(a) Velocity is the rate of change of displacement with time.	1/2
Drift velocity is the average velocity of free electrons under the influence of external electric field.	1/2
(b) $I = neAv_d$	1/2
$A = \frac{I}{nev_d}$	
$A = \frac{3.4}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 0.2 \times 10^{-3}}$	
$A = 1.25 \times 10^{-6} \text{ m}^2$	1/2

22. (a) Define the term 'drift velocity' of conduction electrons in a conductor.
- (b) A conductor of length l and area of cross-section A is connected across an ideal battery of emf V . Derive the formula for the current density in terms of relaxation time τ .

3

(a) Drift velocity is the average velocity with which the free electrons move under external electric field in a conductor.

1

(b) Current density

$$j = \frac{I}{A}$$

$\frac{1}{2}$

$$\therefore I = neAv_d$$

$\frac{1}{2}$

$$\therefore j = nev_d$$

$$\text{But } v_d = \frac{eE\tau}{m}$$

$\frac{1}{2}$

$$\therefore j = \frac{ne^2\tau E}{m} = \frac{ne^2\tau}{m} \left(\frac{V}{l} \right)$$

$\frac{1}{2}$

17. A 20.0 cm long wire is connected across an ideal battery of 4 V. The drift velocity of conduction electrons in the wire is 0.8 mm/s. Calculate the relaxation time of the electrons.

17. A 20.0 cm long wire is connected across an ideal battery of 4 V. The drift velocity of conduction electrons in the wire is 0.8 mm/s. Calculate the relaxation time of the electrons. 2

$$\text{Drift velocity, } v_d = \frac{eE\tau}{m}$$

$$\text{using } E = \frac{V}{\ell} \quad \text{and} \quad \tau = \frac{mv_d \ell}{eV}$$

$$\tau = \frac{0.8 \times 10^{-3} \times 9.1 \times 10^{-31} \times 0.20}{1.6 \times 10^{-19} \times 4}$$

$$\tau = 2.28 \times 10^{-16} \text{ s}$$

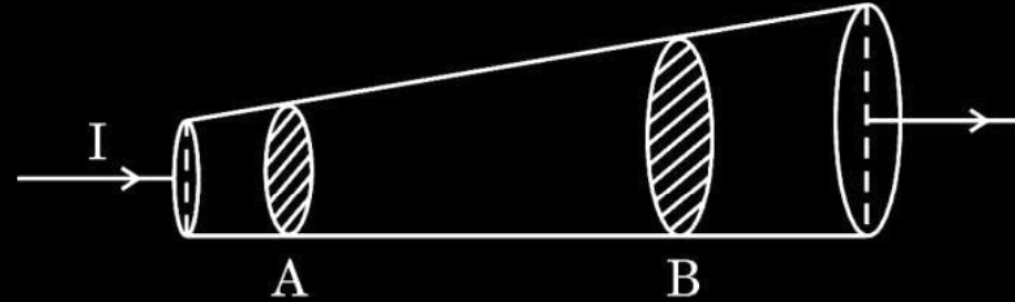
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) Current I ($= 1 \text{ A}$) is passing through a copper rod ($n = 8.5 \times 10^{28} \text{ m}^{-3}$) of varying cross-sections as shown in the figure. The areas of cross-section at points A and B along its length are $1.0 \times 10^{-7} \text{ m}^2$ and $2.0 \times 10^{-7} \text{ m}^2$ respectively. Calculate :



- (I) the ratio of electric fields at points A and B.
- (II) the drift velocity of free electrons at point B.

(i) Calculating	
(I) ratio of electric fields at points A & B	1 ½
(II) drift velocity of free electrons at point B	1 ½
(ii) Finding net electric field at point $\vec{r}$	2

(i)	(I)	$\vec{j} = \sigma \vec{E}$	$\frac{1}{2}$
-----	-----	----------------------------	---------------

$$\frac{j_A}{j_B} = \frac{E_A}{E_B}$$

$$= \frac{I/A_A}{I/A_B}$$

$$= \frac{A_B}{A_A}$$

$$= \frac{2}{1}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(II)	$v_d = \frac{I}{neA}$	$\frac{1}{2}$
------	-----------------------	---------------

$$= \frac{1}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 2 \times 10^{-7}}$$

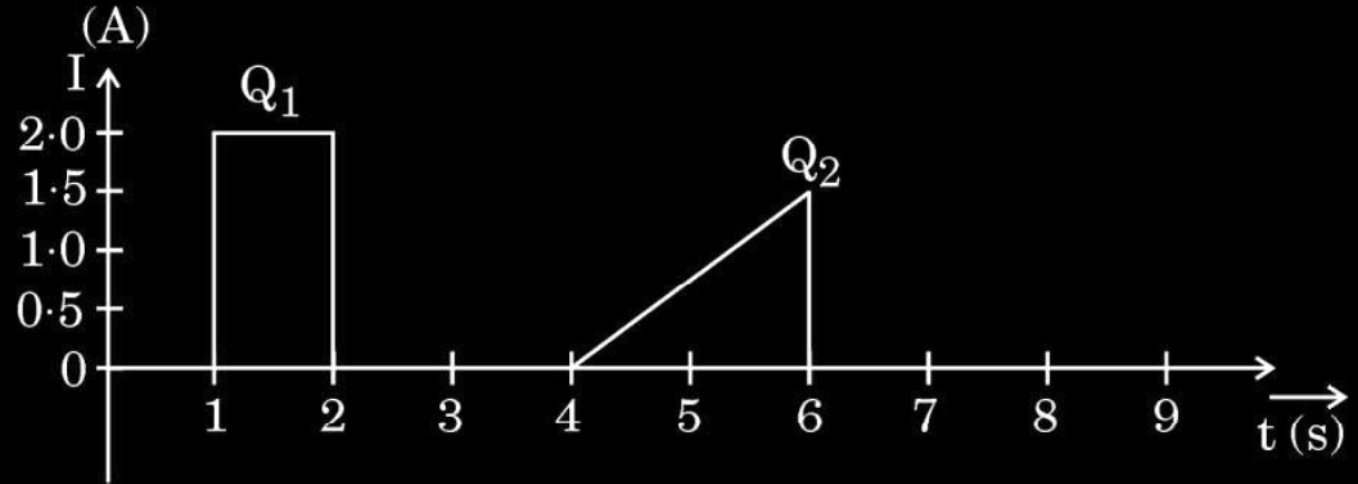
$$= 3.6 \times 10^{-4} \text{ m/s}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

27. (a) (i) Derive an expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.
- (ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals.

3

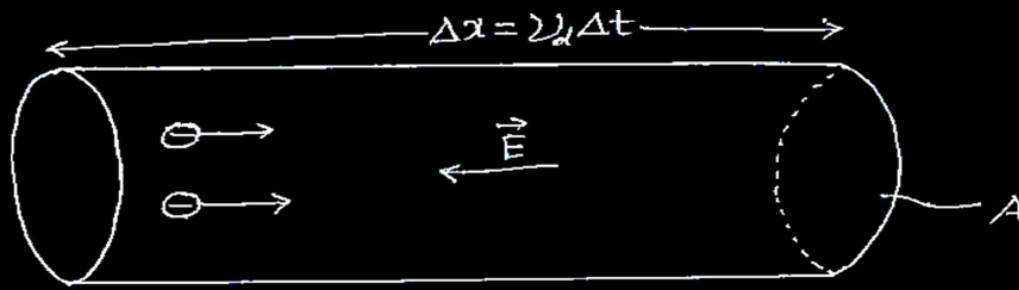


(i) Deriving the expression for resistivity of a conductor

2

(ii) Comparison of charges Q_1 and Q_2

1



Total charge transported along E is

$$I \Delta t = \frac{e^2 A}{m} \tau n \Delta t E$$

$$\frac{I}{A} = \frac{ne^2}{m} \tau E$$

$$J = \frac{1}{\rho} E$$

$$\rho = \frac{m}{ne^2 \tau}$$

(ii) From given graph

$$\frac{Q_1}{Q_2} = \frac{A_1 (\text{Area of rectangle})}{A_2 (\text{Area of triangle})}$$

$$\frac{Q_1}{Q_2} = \frac{2}{3/2}$$

$$Q_1 > Q_2$$

1/2

1/2

1/2

1/2

1/2

1/2

22. (a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of free electrons.
- (b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).

3

(a) Defining Electrical conductivity	1
Obtaining expression of electrical conductivity	1
(b) Explaining qualitative change in resistivity with temperature	1

(a) Electrical conductivity: - It is the reciprocal of the resistivity.

Alternatively: -

$$\sigma = 1 / \rho$$

$$I_{\Delta t} = + neA |\vec{v}_d|_{\Delta t}$$

$$\text{Substituting } |\vec{v}_d| = \frac{e|\vec{E}|}{m} \tau$$

$$I_{\Delta t} = \frac{e^2 A}{m} \tau n_{\Delta t} |\vec{E}|$$

$$I = |\vec{j}| A$$

$$\Rightarrow |\vec{j}| = \frac{ne^2}{m} \tau |\vec{E}| \dots \dots \dots (1)$$

$$\vec{j} = \sigma \vec{E} \dots \dots \dots (2)$$

Comparing eq.(1) & (2):-

$$\sigma = \frac{ne^2}{m} \tau$$

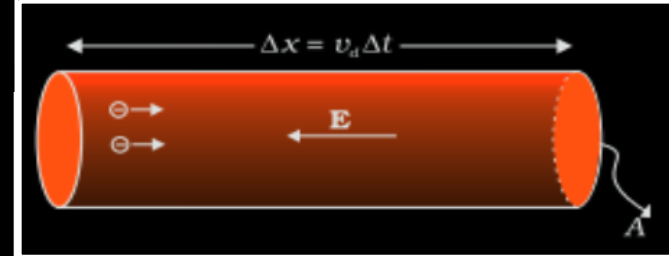
(b) On increasing the temperature, the value of τ decreases as a consequence conductivity decreases and hence resistivity increases.

1

1/2

1/2

1



2. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be -

- (A) 2 (B) $\frac{1}{2}$
(C) $\frac{3}{2}$ (D) $\frac{8}{9}$

2. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be -

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$

(D) $\frac{8}{9}$

- (b) (i) A conductor of length l is connected across an ideal cell of emf E . Keeping the cell connected, the length of the conductor is increased to $2l$ by gradually stretching it. If R and R' are initial and final values of resistance and v_d and v_d' are initial and final values of drift velocity, find the relation between (i) R' and R and (ii) v_d' and v_d .
- (ii) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction ?

(i) $l' = 2l$
 $Al = A'l' = \text{volume of the wire}$
 $Al = A'(2l)$
 $\frac{A}{2} = A'$
 $R = \frac{\rho l}{A}$
 $R' = \frac{\rho l'}{A'}$
 $R' = \frac{\rho(2l)}{A/2}$
 $\frac{R'}{R} = 4$

Alternatively

$R' = n^2 R$
 $n = 2$
 $R' = 4R$
(ii) $v_d = \frac{eE}{m} \tau$
 $v_d = \frac{eV}{ml} \tau$
 $v_d' = \frac{eV}{ml'} \tau$
 $\frac{v_d'}{v_d} = \frac{l}{l'} = \frac{1}{2}$

$\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$

(ii) No

12. A copper wire of non-uniform area of cross-section is connected to a d.c. battery. The physical quantity which remains constant along the wire is _____

12. A copper wire of non-uniform area of cross-section is connected to a d.c. battery. The physical quantity which remains constant along the wire is Electric current

5. $\text{m}^2\text{V}^{-1}\text{s}^{-1}$ is the SI unit of which of the following ?

1

- (A) Drift velocity
- (B) Mobility
- (C) Resistivity
- (D) Potential gradient

5. $\text{m}^2\text{V}^{-1}\text{s}^{-1}$ is the SI unit of which of the following ?

1

(A) Drift velocity

(B) Mobility

(C) Resistivity

(D) Potential gradient

28. (a) Differentiate between the random velocity and the drift velocity of electrons in an electrical conductor. Give their order of magnitudes.
- (b) A conductor of uniform cross-sectional area is connected across a dc source of variable voltage. Draw a graph showing variation of drift velocity of electrons (v_d) as a function of current density (J) in it.

3

a) differentiating between random velocity and drift velocity 1 mark

Order of magnitude 1 mark

b) drawing the graph showing the variation of drift velocity as a function of Current density 1 mark

Random Velocity v	Drift Velocity v_d
1. The velocity acquired by the free electrons in the absence of electric field.	1. The average velocity acquired by the free electrons in presence of electric field.
2. The average random velocity is zero.	2. The average drift velocity is not zero.
3. Has quite a large value	3. Has a very small value

1

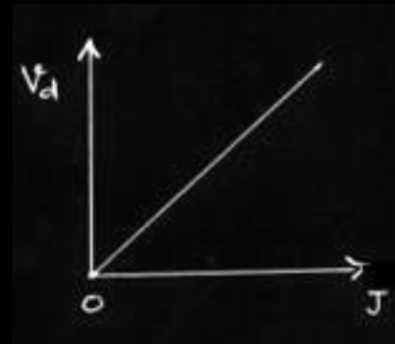
Order of magnitude of random velocity is 10^2m/s .

Order of magnitude of drift velocity is 10^{-3}m/s .

$\frac{1}{2}$

$\frac{1}{2}$

[Note: If the student writes drift speed is nearly 10^{-5} times smaller than random velocity ,award the last 1 mark]



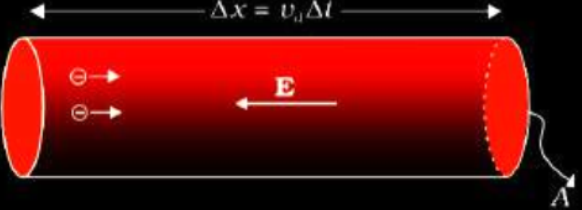
1

[if a student writes $J = \frac{I}{A} = \frac{n e A v_d}{A} = n e v_d$ but does not draw the graph award $\frac{1}{2}$ mark only]

- 21.** Define the term 'mobility' of charge carriers in a current carrying conductor. Obtain the relation for mobility in terms of relaxation time. 2

OR

Define the term 'drift velocity' of electrons in a current carrying conductor. Obtain the relationship between the current density and the drift velocity of electrons. 2

Mobility is defined as the magnitude of drift velocity per unit electric field. $\mu = \frac{ \vec{V}_d }{E}$ [Even if a student writes only the mathematical relation award ½ mark] Given $V_d = \frac{e\tau E}{m}$ Hence, $\mu = \frac{V_d}{E} = \frac{e\tau}{m}$	1 ½ ½
The average speed with which electrons move when an electric field or potential difference is applied is called drift velocity. $\vec{V}_d = \frac{-e\vec{E}\tau}{m}$ [Award 1/2mark if student writes the formulae]  The amount of charge crossing the area A in time Δt $I\Delta t = ne A \vec{V}_d \Delta t$ Hence current density $j = \frac{I}{A} = ne V_d$	1 ½ ½

How does the mobility of electrons in a conductor change, if the potential difference applied across the conductor is doubled, keeping the length and temperature of the conductor constant ?

How does the mobility of electrons in a conductor change, if the potential difference applied across the conductor is doubled, keeping the length and temperature of the conductor constant ?

No Change	1
-----------	---

- (i) Derive an expression for drift velocity of electrons in a conductor. Hence deduce Ohm's law.
- (ii) A wire whose cross-sectional area is increasing linearly from its one end to the other, is connected across a battery of V volts. Which of the following quantities remain constant in the wire ?
- (a) drift speed
 - (b) current density
 - (c) electric current
 - (d) electric field

Justify your answer.

Let an electric field E be applied the conductor. Acceleration of each electron is

$$a = -\frac{eE}{m}$$

Velocity gained by the electron

$$v = -\frac{eE}{m}t$$

Let the conductor contain n electrons per unit volume. The average value of time ' t ', between their successive collisions, is the relaxation time, ' τ '.

Hence average drift velocity $v_d = \frac{-eE}{m} \tau$

The amount of charge, crossing area A , in time Δt , is

$$\equiv neAv_d\Delta t = I\Delta t$$

Substituting the value of v_d , we get

$$I\Delta t = neA\left(\frac{eE\tau}{m}\right)\Delta t$$

$$\therefore I = \left(\frac{e^2A\tau n}{m}\right)E = \sigma E, \left(\sigma = \frac{e^2\tau n}{m} \text{ is the conductivity}\right)$$

But $I = JA$, where J is the current density

$$\Rightarrow J = \left(\frac{e^2\tau n}{m}\right)E$$

$$\Rightarrow J = \sigma E$$

This is Ohm's law

[Note : Credit should be given if the student derives the alternative form of Ohm's law by substituting $E = \frac{V}{\ell}$]

ii) Electric current will remain constant in the wire.

All other quantities, depend on the cross sectional area of the wire.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (i) Define the term drift velocity.
- (ii) On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend ?
- (iii) Why alloys like constantan and manganin are used for making standard resistors ?

i) Average velocity acquired by the electrons in the conductor in the presence of external electric field.

1

[**Alternatively:**

$$v_d = \frac{-eE\tau}{m} \text{ where } \tau \text{ is the relaxation time.}]$$

ii) $v_d = \frac{-eE\tau}{m}$

We have $E = -\frac{V}{\ell}$, where V is potential difference across the length ' ℓ ' of the conductor

$$v_d = \frac{eV\tau}{m\ell}$$

1/2

Current flowing $I = neAv_d$

1/2

$$I = neAv_d \frac{eV\tau}{m\ell} = \frac{ne^2AV\tau}{m\ell}$$

1/2

$$\frac{I}{V} = \frac{ne^2A\tau}{m\ell} = \frac{1}{R} \text{ -----(i)}$$

Also, $R = \rho \frac{\ell}{A}$ -----(ii)

Comparing (i) and (ii)

$$\rho = \frac{m}{ne^2\tau}$$

1/2

Resistivity of the material of a conductor depends on the relaxation time, i.e., temperature and the number density of electrons.

1/2 + 1/2

iii) Because constantan and manganin show very weak dependence of resistivity on temperature

1

Define mobility of a charge carrier. What is its relation with relaxation time ?

When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, calculate the resistivity of the material of wire.

When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, calculate the resistivity of the material of wire.

$$R = \rho \frac{l}{A}; I = neAv_d$$

1/2

$$\therefore \rho = \frac{V}{nelv_d}$$

1/2

Alternatively,

$$\left(j = \sigma E = \frac{E}{\rho} \text{ or } \frac{E}{j} = \rho \right)$$

$$\therefore \rho = \frac{V}{lnelv_d}$$

(Award this 1 mark even if the student writes the formula for ρ directly as such)

$$\therefore \rho = \frac{5}{0.1 \times 8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4}} \Omega - m$$

$$= 1.56 \times 10^{-5} \Omega - m$$

$$\simeq 1.6 \times 10^{-5} \Omega - m$$

1/2

1/2

- (i) Derive an expression for drift velocity of free electrons.
- (ii) How does drift velocity of electrons in a metallic conductor vary with increase in temperature ? Explain.

- i. When a potential difference is applied across a conductor, an electric field is produced and free electrons are acted upon by an electric force ($= -Ee$). Due to this, electrons accelerate and keep colliding with each other and acquire a constant (average) velocity v_d

$\frac{1}{2}$

$$\therefore, F_e = -Ee$$

$$\therefore, F_e = \frac{-eV}{l}$$

$\frac{1}{2}$

$$\text{As } a = \frac{-F}{m} = \frac{-eV}{m}$$

$$\text{as } v = u + at$$

$$u = 0, t = \tau \text{ (relaxation time)}$$

$\frac{1}{2}$

$$v_d = -a \tau$$

$$v_d = \frac{-eV}{lm} \tau$$

$\frac{1}{2}$

- ii. Decreases, as time of relaxation decreases.

$\frac{1}{2}, \frac{1}{2}$